

# Computer Networking: Protocols, OSI Reference Model & Architectures

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## 1. The 7-Layer OSI Reference Model Comparison

The Open Systems Interconnection (OSI) reference model organizes network communication into 7 distinct abstraction layers. The following table provides a comprehensive technical comparison across all layers:

| Layer # | Layer Name   | Protocol Data Unit (PDU) | Primary Addressing     | Key Protocols           | Hardware / Devices      |
|---------|--------------|--------------------------|------------------------|-------------------------|-------------------------|
| 7       | Application  | Data / Message           | Names / URLs / URIs    | HTTP, DNS, SMTP, SSH    | Gateways, Firewalls     |
| 6       | Presentation | Data                     | Encoding Syntax        | TLS/SSL, ASCII, JPEG    | Software Libraries      |
| 5       | Session      | Data                     | Session IDs, Tokens    | RPC, NetBIOS, PPTP      | OS Sockets, Middleware  |
| 4       | Transport    | Segment (TCP) / Datagram | Port Numbers (16-bit)  | TCP, UDP, SCTP, QUIC    | L4 Load Balancers       |
| 3       | Network      | Packet                   | IP Address (IPv4/IPv6) | IPv4, IPv6, ICMP, BGP   | Routers, L3 Switches    |
| 2       | Data Link    | Frame                    | MAC Address (48-bit)   | Ethernet (802.3), Wi-Fi | Switches, Bridges, NICs |
| 1       | Physical     | Bit Stream / Signal      | Frequency / Voltage    | 1000BASE-T, RS-232      | Hubs, Repeaters, Cables |

## 2. Transport Layer: TCP vs UDP Technical Specification

The Transport Layer delivers end-to-end communication services. Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) represent contrasting transport philosophies:

| Feature / Metric          | Transmission Control Protocol (TCP)                         | User Datagram Protocol (UDP)                             |
|---------------------------|-------------------------------------------------------------|----------------------------------------------------------|
| Connection State          | Connection-Oriented (3-way handshake SYN, SYN-ACK, ACK)     | Connectionless (No pre-communication handshake)          |
| Reliability Guarantee     | Guaranteed delivery with sequence numbering and ACK         | Best-effort delivery; packet loss possible               |
| Ordering Guarantee        | Strict in-order packet delivery using byte sequence numbers | No ordering guarantee; packets may arrive out-of-order   |
| Flow & Congestion Control | Sliding window flow control, Reno/Cubic congestion control  | None; application must handle congestion throttling      |
| Header Size Overhead      | 20 bytes minimum, up to 60 bytes with options               | Fixed 8 bytes (Source Port, Dest Port, Length, Checksum) |
| Transmission Mode         | Byte stream abstraction; segments delimited by window       | Discrete message datagram boundaries preserved           |
| Primary Use Cases         | Web browsing (HTTP/HTTPS), File transfers (FTP), SSH, Email | Video streaming (RTP), DNS queries, Voice over IP (VoIP) |

## 3. Well-Known Service Port Allocations

Standard port numbers defined by IANA for network infrastructure services include: Port 20/21 (FTP), Port 22 (SSH), Port 25 (SMTP), Port 53 (DNS - UDP/TCP), Port 67/68 (DHCP), Port 80 (HTTP), Port 110 (POP3), Port 143 (IMAP), Port 443 (HTTPS), and Port 3306 (MySQL).

## 4. Network Layer: Dynamic Routing Protocols Comparison

Routing protocols compute optimal forwarding paths across complex autonomous network topologies:

| Routing Protocol | Type / Category        | Algorithm Basis                 | Metric Formula                                | Convergence Speed                   | Administrative Weight   |
|------------------|------------------------|---------------------------------|-----------------------------------------------|-------------------------------------|-------------------------|
| RIPv2            | Interior Gateway (IGP) | Distance Vector (Bellman-Ford)  | Hop Count (Max 15 hops)                       | Slow (count-to-infinity)            | 120                     |
| OSPFv2 / v3      | Interior Gateway (IGP) | Link State (Dijkstra SPF)       | Cost = 10^8 / Bandwidth                       | Fast (hierarchical areas)           | 110                     |
| IS-IS            | Interior Gateway (IGP) | Link State (Dijkstra SPF)       | Interface Metric (default 10)                 | Very Fast (sub-second)              | 115                     |
| EIGRP            | Interior Gateway (IGP) | Advanced Distance Vector (DUAL) | Bandwidth + Delay composite                   | Instantaneous with Feasibility      | 90 (Internal)           |
| BGP-4            | Exterior Gateway (EGP) | Path Vector (AS Path lists)     | Policy attributes (Local Pref, AS-Path, etc.) | Variable (designed for scalability) | 200 (iBGP) / 200 (iBGP) |

## 5. Border Gateway Protocol (BGP) Mechanics

BGP is the path-vector routing protocol running the global core Internet. Unlike interior gateway protocols that minimize link cost, BGP routes based on autonomous system (AS) path vectors and organizational peering agreements. A BGP router establishes TCP sessions over port 179 to advertise prefix reachability through BGP UPDATE messages.

## 6. Internet Protocol Header Formats & Addressing

The network layer encapsulates transport segments into IP packets. The following table contrasts the IPv4 and IPv6 header architectures:

| Header Attribute     | Internet Protocol version 4 (IPv4)                        | Internet Protocol version 6 (IPv6)                         |
|----------------------|-----------------------------------------------------------|------------------------------------------------------------|
| Address Length       | 32 bits (4 bytes); ~4.29 billion unique addresses         | 128 bits (16 bytes); ~3.4 x 10^38 unique addresses         |
| Base Header Size     | 20 bytes minimum; variable with optional options field    | Fixed 40 bytes; extensions linked via Next Header pointer  |
| Checksum Validation  | Header Checksum verified at every intermediate hop router | No header checksum; error detection delegated to L2 and L4 |
| Packet Fragmentation | Allowed by intermediate routers and source endpoints      | Performed exclusively by the sending source endpoint       |
| Configuration Model  | Manual static assignment or DHCP server delegation        | Stateless Address Autoconfiguration (SLAAC) or DHCPv6      |
| Packet Lifetime      | Time to Live (TTL) decremented by 1 per hop router        | Hop Limit field decremented by 1 per hop router            |

## 7. Address Resolution Protocol (ARP) & ICMP

ARP maps logical 32-bit IP addresses to physical 48-bit Ethernet MAC addresses within a local broadcast domain. When a host intends to transmit to an IP on the subnet without a cached MAC, it broadcasts an ARP Request (Who has 192.168.1.1? Tell 192.168.1.50). Internet Control Message Protocol (ICMP) reports network diagnostic conditions and routing errors, powering ping (Echo Request/Reply) and traceroute.